Status	Finished
Started	Saturday, 10 October 2026, 6:17 PM
Completed	Saturday, 10 October 2026, 6:23 PM
Duration	6 mins 10 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Dividing the input into parts, solving each part, and combining the solutions
b. Rearranging data without sorting
c. Sorting data sequentially
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Merge Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It bubbles up the largest element in each pass
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Bubbling up the largest element to its correct position with each pass
c. Being the most efficient sorting algorithm
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It allows for efficient parallel processing and sorting of data
c. It only works on small datasets
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It only works with integer arrays
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Insertion
b. Distribution
c. Extraction
d. Selection

The correct answer is:
Insertion

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. Searching
c. All of the above
d. Duplicates

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes the data unsorted
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **15**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is difficult to implement
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists